

Q19 From Fig. 19.1, when the current is 5 mA, the p.d. across the diode is 0.8 V

The p.d. across the 50 Ω resistor is given by $V = (5 \text{ mA})(50 \Omega) = 0.25 \text{ V}$

So the p.d. across the supply = $0.8 + 0.25 = 1.05 \text{ V}$

Answer: C

Q20 Potential at X is 12 V